

Ayush Girhay

B.Tech. III Year, Civil Engineering
Indian Institute of Technology Roorkee

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Profile

AI/ML and Full Stack Developer with experience building scalable web applications, AI-powered automation workflows, and machine learning solutions using Python, React, Node.js, and Generative AI technologies.

AIR-6707 JEE Advanced 2023 **AIR-1283** JEE Mains 2023

Experience

TRS Consultant May 2026 – JUL 2026

AI & Full Stack Developer Intern, contributing to client projects by developing full-stack web applications, integrating AI-powered features, building REST APIs, automating business workflows, and delivering scalable digital solutions to improve operational efficiency.

Curious About Sales Feb 2026 – Apr 2026

AI Intern, engineered AI-enabled automation workflows using **n8n**, developed lead enrichment and qualification pipelines, integrated third-party sales platforms and APIs, and automated outreach processes to improve operational efficiency.

Technical Skills

Languages: Python, C++, SQL, HTML, CSS, JavaScript, Bootstrap, React.js, ROS2

Data Analytics: EDA, Pandas, NumPy, Matplotlib, Data Visualization

Software Packages: ArcGIS, QGIS, Solidworks, Fusion360

Tools: Jupyter Notebook, Google Colab, scikit-learn, MySQL, Tableau, Excel, Git, GitHub, VS Code, Postman

Projects

Creating a Large Language Model Jun 2026 – Present

Built a decoder-only LLM from scratch in Python by implementing tokenization, embeddings, transformer blocks, self-attention, and autoregressive text generation with a complete training pipeline.

EventSphere – AI-Powered Event and Media Management Platform May 2026 – Jun 2026

Developed a full-stack event management platform using React, Node.js, Express, and MongoDB featuring authentication, ticket booking, QR attendance, media sharing, and role-based access control.

Dynamic Tariff Optimization for EV Charging Networks Sep 2023 – Nov 2023

Built an AI-driven pricing engine for EV charging stations using demand forecasting and occupancy prediction to optimize tariffs and improve resource utilization.

Dynamic Pricing for Urban Parking Lots May 2025 – Jul 2025

Developed a real-time parking pricing system using Python, Pandas, NumPy, Pathway, and Bokeh by analyzing demand, traffic, occupancy, and competitor pricing across 14 parking lots.

House Price Prediction Dec 2025 – Jan 2026

Developed a multimodal ML model combining tabular data and satellite imagery using XGBoost, PCA, SHAP, and Grad-CAM for accurate and explainable house price prediction.

Positions of Responsibility

Captain — Hockey, IIT Roorkee Jul 2026 – Present

Led the IIT Roorkee hockey team, coordinating training, mentoring players, and representing the institute in tournaments while strengthening leadership, teamwork, and strategic decision-making.

Core Member — Models and Robotics Section (MARS), IIT Roorkee Jun 2025 – Present

Contributing to robotics projects involving CAD design, AI/ML integration, and autonomous systems using SolidWorks and Fusion 360.

Education

Indian Institute of Technology Roorkee 2023 – Present

B.Tech. in Civil Engineering

Insight Junior College (MSBSHSE) 2021 – 2022

Class XII

St. Xavier High School (CBSE) 2019 – 2020

Class X